AJEENKYA D Y PATIL UNIVERSITY | School of Engineering

Smart Examination Management System

AI-POWERED SEATING ALLOCATION • ANTI-CHEATING • REAL-TIME MONITORING

PROBLEM STATEMENT

Manual examination seating allocation is labor-intensive and prone to human errors during large-scale exams. The traditional approach requires administrators to manually place hundreds of students, which consumes significant time and resources.

Predictable seating arrangements increase the probability of cheating as students from the same class tend to be seated near each other. Without algorithmic constraints, same-class students may end up on the same bench, making it easier to copy or communicate during the examination.

Traditional invigilation lacks real-time monitoring capabilities, making it difficult to detect suspicious activities or manage examination halls efficiently. There is no automated system that ensures fair and risk-aware seat distribution across the examination venue.

Inefficient bulk student data management through spreadsheets leads to data entry errors and makes it challenging to maintain accurate records. The lack of integration between student databases and seating allocation systems creates additional operational overhead.

OBJECTIVES

Develop an automated seating allocation system using heuristic optimization algorithms that can process hundreds of students within seconds. The system should minimize human intervention while maximizing seating efficiency.

Implement risk-based seat distribution to minimize cheating probability by ensuring students from the same class are never seated on the same bench. The algorithm calculates risk scores for each potential seat assignment and optimizes accordingly.

Create a real-time monitoring dashboard for invigilators that provides live insights into seating arrangements, student分布, and examination hall status. This enables quick decision-making during examination emergencies.

Enable bulk student data upload with Excel file processing to streamline the onboarding process. The system parses Excel files, validates data, and automatically classifies students by their respective classes.

Provide anti-cheat bench separation based on class constraints, ensuring that no two students from the same class share a bench regardless of row placement.

ALGORITHM THEORY

Risk Minimization Approach

Constraint Satisfaction: The algorithm enforces the fundamental constraint that students from the same class cannot be seated on the same bench. This is achieved through a hard constraint in the allocation logic.

Risk Scoring: Each potential seat assignment is assigned a risk score based on the proximity of same-class students in surrounding seats. Lower scores indicate safer arrangements.

Greedy Optimization: Seats are allocated iteratively, with each selection choosing the lowest-risk available seat at that moment. This approach provides near-optimal solutions efficiently.

Gap Pattern: Strategic gaps are inserted at regular intervals to break potential communication chains between students from different but related classes.

WORKFLOW

Data Input → Preprocess → Risk Matrix

Algorithm → Allocate → Export

METHODOLOGY

Data Modeling: MongoDB is used with flexible schemas to store student records, classroom configurations, and seating plans. The document-based model allows easy scaling and modification of data structures.

Preprocessing Pipeline: The system parses Excel files containing student data, validates entries for completeness and correctness, and automatically extracts class information for categorization.

Risk Engine: A mathematical formulation calculates seating risk based on class adjacency. The engine considers bench position, row distance, and class proximity to generate comprehensive risk scores.

Optimization: A greedy algorithm with backtracking iterates through students, placing each in the lowest-risk available seat while maintaining all constraints. The approach balances optimality with computational efficiency.

Anti-Cheat Logic: The bench-level separation constraint ensures complete isolation of same-class students. The algorithm validates each allocation against this hard constraint before confirmation.

SYSTEM ARCHITECTURE

Flutter App → REST API → Node.js → MongoDB

Express.js | Flutter | MongoDB | JWT Auth | Multer | Mongoose

KEY RESULTS

70%	100%	0%
Faster Allocation	Automated	Same-Bench Risk

LIMITATIONS

The system requires structured data input in a specific format for processing. Excel files must follow the defined template with proper column headers.

Currently operates in single exam hall mode, which may require duplication for multi-hall examinations. Performance is hardware dependent on the server infrastructure.

CONCLUSION

This smart automated examination system significantly reduces manual workload by eliminating hours of manual seating arrangement. The AI-driven risk minimization approach ensures fairness and academic integrity throughout the examination process.

The scalable architecture is designed for future expansion to multi-hall deployment, supporting institutions as they grow. The comprehensive audit trail provides examination transparency for administrators and regulatory bodies.

By integrating modern web technologies with optimization algorithms, this system represents a significant advancement in examination management, setting new standards for efficiency and fairness in academic institutions.

TEAM MEMBERS

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PROJECT GUIDE

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